

CBOE Global Markets CBOE

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by

GLSV

2020 2021

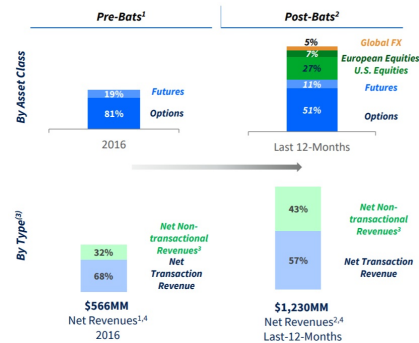
Price:	86.84	EPS	5.25	5.04
Shares Out. (in M):	109	P/E	16.5	17.2
Market Cap (in \$M):	9,518	P/FCF	14.5	15
Net Debt (in \$M):	502	EBIT	698	672
TEV (in \$M):	10,019	TEV/EBIT	14.3	14.8

Description

CBOE is the holding company for the Chicago Board Options Exchange, one of the world's largest options exchanges. The company offers a number of different products and services for risk managers across the globe, with exclusive rights to options of the S&P 500 and the VIX product suite. For most of the company's history, CBOE focused on the trading of US options and futures. After the acquisition of BATS in 2017, CBOE expanded into US and European equities trading and began reporting five business segments (% of LTM revenues): Options (51%), U.S. Equities (27%), Futures (11%), European Equities (7%), and Global FX (5%). Supersny provided a thorough [write-up](#) on VIC in 2018 which walks through the history of the company, management, and the industry background. In brief, CBOE operates in an oligopolistic industry, with high margins, meaningful barriers to entry, and a scalable business model.

Similar to the conclusion at that time (note the stock is actually cheaper today), we believe the market is currently giving investors the opportunity to own a very high quality franchise at a discounted valuation thereby providing asymmetric return potential. Similar to the aftermath of the volatility spike in early 2018, investors appear to be concerned about the growth potential of the core VIX and S&P franchises (note 2Q-3Q 2020 volumes have been soft) and potential contribution from new products. We address those questions below:

Figure 1 -- Revenue Breakdown



What is the outlook for the core VIX product set and the SPX Options business?

CBOE created the VIX, commonly known as the "fear index," which represents the market's expectation of forward 30-day volatility, and launched the corresponding futures and options contracts in the mid-2000s. Because the VIX is a proprietary product of CBOE, investors can trade its options and futures only through CBOE. While the term "volatility" leads people to connect the VIX products with unpredictability, the long-term growth rates of these products have been remarkably strong. The CAGR of VIX options ADV is approximately 27% since 2006 and that of VIX futures ADV is roughly 23% since 2011. Besides the VIX product suite, CBOE also has exclusive products in the options of S&P indices. Back in 2013, CBOE renewed its license with S&P Global which allows CBOE to exclusively trade the index options of the S&P 500, S&P 100 and S&P Select Sector until December 31, 2032 and non-exclusively in 2033. This agreement is indicative of the CBOE's market presence and influence and helps explain the approximately 11% CAGR of SPX Options ADV since 2005. Finally, while the contributions are small, CBOE also signed exclusive agreements with MSCI and FTSE Russell for the index options. While the growth rates mentioned above are key to CBOE's consistent financial performance, the company will need to maintain or grow share in these core businesses moving forward, while diversifying into more product offerings.

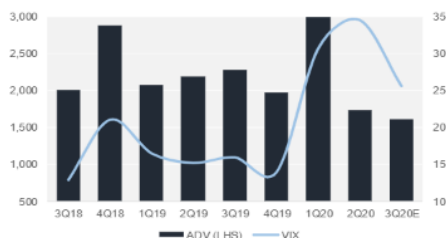
With respect to the VIX products, there are a few things to highlight. First, VIX futures are the most effective when the VIX curve is positively sloped, the VVIX (vol of vol) is high and the VIX itself is low. In other words, there are a lot of moving pieces that need to be aligned to give VIX futures an ideal operating environment. Second, the number of participants actively trading VIX futures is narrow and can be volatile. For example, in 2018 when volatility spiked, the number of participants in the market hit a record but quickly fell once vol normalized, ending the year below where it started. Earlier this year, following the spike in vol in March, participation fell to a multi-year low, but it has rebounded to long-term averages since. Finally, VIX futures ADV is highly correlated to volatility ETPs, with more AUM and greater inflows at these ETPs driving better demand for VIX futures. These 3 factors can lead to some choppiness in demand for VIX futures and weigh on Open Interest (OI), a precursor to future volumes. A few things to consider on OI and demand. Several market participants use multiple CBOE products, including SPX puts and VIX options in order to hedge their activities. So while some investors might exit futures contracts, they are more often than not simply migrating to another CBOE product. In terms of ETPs and demand, we would point to the discontinuation of TVIX, which on the surface should have reduced demand for VIX futures. In the first month alone since the discontinuation, alternative products (VXX, UVXY, VIXY) had more inflows on a combined basis than TVIX had outflows. Since discontinuation, the AUM in all volatility ETPs has exceeded \$4B, above the pre-COVID levels (excluding the major spike seen during the shutdown). That said, OI is down significantly from the spikes seen earlier this year but off of the lows at this point, as the exchange was closed until July.

Moving to VIX and SPX options, the concern is the competitive landscape as it pertains to CME's E-mini products and the retail usage of SPY options. While SPX options have been

heavily institutionally focused, they have been gaining some traction with retail investors, but the SPY puts/calls are much more “retail friendly.” At the same time, institutions are getting more comfortable with the E-minis product suite, which could start to erode market share. VIX and SPX options OI looks very similar to the futures OI, bouncing off the lows but well below the COVID spike levels. CBOE carries roughly a 36% market share in options, with a 99.4% share in index and 32% share in multi-listed (both up YoY in Q220). Therefore, while the competitive landscape is and always has been tough, CBOE has grown ADV consistently for many years and this should be the case moving forward.

The proprietary product suite (which accounts for over 60% of TTM trading revenues) continues to track lower in 3Q as shown in Figure 2. Management has attributed the recent slowdown to uncertainty around the pandemic, which has kept institutional investors on the sidelines and depressed overall product usage. Looking ahead, there are three items that should help volumes rebound from current levels, including 1) VIX levels now in the low 20s, 2) steepening VIX curve, 3) new product roll out including retail friendly Mini VIX futures which has seen solid volumes since its August launch.

Figure 2 -- Proprietary Products Quarterly ADV

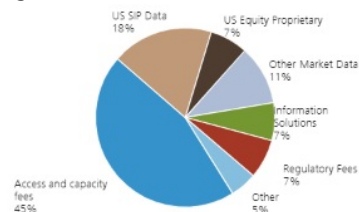


Can new product development and recent acquisitions move the revenue/earnings needle?

Prior to the acquisition of BATS in March 2017, CBOE mainly offered proprietary options and futures trading products. These products represented 100% of transaction fees and totaled approximately 72% of revenues. With BATS, CBOE added new product lines in U.S. Equities, European Equities and Global FX, as well as expanded capabilities in market data and listing venues. In the front end, the combined entity offers a variety of pricing and market models. In the back end, CBOE utilizes BATS’s sophisticated technology to achieve cost savings. The CBOE had been diversifying away from strictly transaction fees for many years, but BATS truly transformed the company away from strictly futures/options, adding significantly to other fee revenues. As a result of the acquisition, revenues have grown nearly 50%, despite a tough transaction fee market in 2019, and EBITDA margins recovered from their initial dip back to the high 60s the last 2 years.

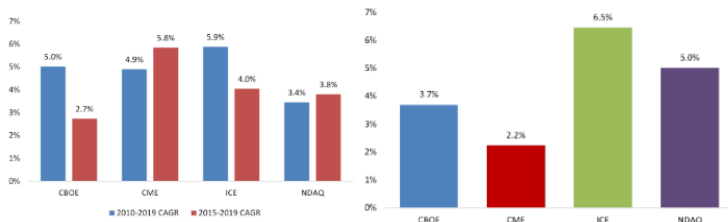
Given current low balance sheet gearing (~1x net debt-to-EBITDA), management has recently been deploying capital on several smaller deals, including EuroCCP and MatchNOW. EuroCCP will extend their European equities business into trading and clearing pan-European derivatives, as well as be the main driver for them to offer European equity index options and futures in the 1H2021. ECCP is expected to be slightly dilutive in the very near-term (2020-21) as management builds out infrastructure and capabilities; however, over time it could become a substantial revenue growth driver (3-5 years) as the European derivatives market is less than 20% the size of the US. With MatchNOW, CBOE gets an alternative trading system (ATS) that carries 7% of the total market in Canada, including 65% market share in dark trading. It is important to note both were cash deals that closed in early Q3, which should be a catalyst for transaction growth in 2021 and beyond. Unlike BATS, which was a game changer, the more recent transactions were more “buy versus build” decisions, which should have fast payoffs and become accretive fairly quickly. In addition to EuroCCP and MatchNOW, CBOE acquired Hanweck, FT Options, and Trade Alert this year in order to build its Information Solutions offering. As previously noted, non-transactional revenue accounts for over 40% of total revenue and should grow at a mid-to-high single digit rate prospectively driven by both organic (e.g. cross selling, new offerings) and inorganic opportunities.

Figure 3 -- Non-Transaction Revenue by Type



In summary, while we acknowledge the CBOE is facing near-term cyclical headwinds, it is also important to note that management has been focusing much of their time & attention on the integration of its transformative acquisition of BATS. Going forward, management plans to “pivot from a massive technology integration” (accomplished without disruption) toward accelerating organic growth by building new technologies including the development of a state-of-the-art research and data platform and enhancing global distribution. Long-term, management has a track record of success driving growth in both its proprietary products and through capital deployment. Per Figures 4, Wall Street is expecting very little in terms of future revenue growth potential which both feeds into the low valuation currently and provides upside if management is successful.

Figure 4 -- Organic Revenue CAGR Figure 5 -- Projected Revenue Growth (2019-2022)

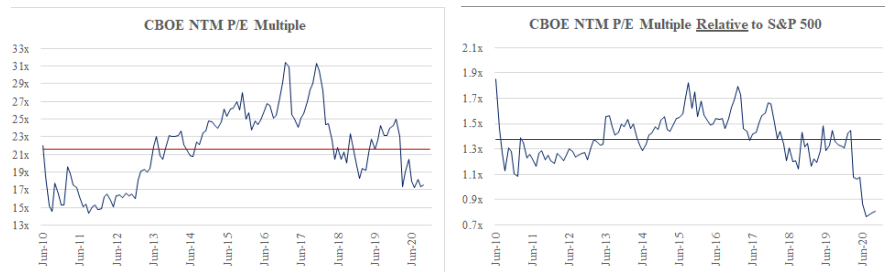


Management

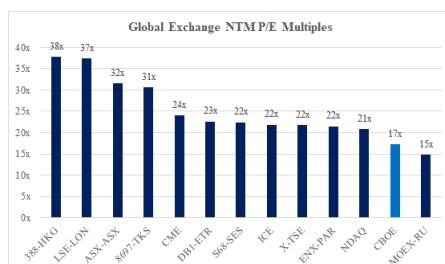
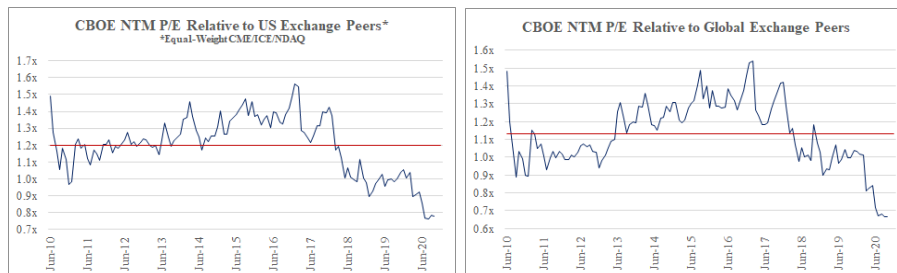
Ed Tilly, is essentially a CBOE “lifer,” having been CEO since 2013 and COO prior to that. The CFO and COO came from BATS, where they were instrumental in growing the company from a nascent exchange into a global force prior to selling to the CBOE. We believe management is very thoughtful in their capital allocation, approach to acquisitions, product pricing, and enhancement of the CBOE brand and product mix.

Valuation

The stock trades at less than 18x forward EPS and 13x forward EBITDA, both of which are a 3 turn discount to exchange peers. Relative to the company’s own trading history, the current valuation is in the bottom quartile, and relative to the broader market the stock trades at a record low valuation. On a trailing free cash flow basis, the stock yields over 7.5% relative to its enterprise value. Based upon these valuation parameters, we see both absolute and relative value in CBOE. We get upside of 25-30% assuming a more normalized valuation multiple based upon mid-\$5 in earnings power (average 2020-2022) and 65% upside over the next few years using the modeling assumptions below which embeds relatively conservative assumptions around growth and capital allocation (e.g. share repurchase). Evaluating the downside risk, assuming the stock trades at a trough multiple of 15x earnings and the company's cash flow merely offsets unexpected and persistent weakness in the underlying business resulting in flat earnings per share, we estimate the stock has about ~10% risk. Last, any political and/or economic volatility ought to be net beneficial to business fundamentals as evidenced by the ~28% surge in revenue in 1Q 2020 amidst the height of the COVID-19 pandemic in the US.



If we look at CBOE’s valuation relative to its exchange peers domestically and globally, the shares are similarly discounted. This stands in contrast to most of the company’s trading history whereby CBOE shares commanded a premium to the industry. This was likely due to an embedded takeover premium as well as the relatively high proportion of revenue generated from futures and exclusively licensed products. The de-rating has occurred despite transaction revenue declining from 90% to ~63% currently and adjusting operating margins expanding to industry-leading levels following the BATS deal. Considering the trend toward ongoing industry consolidation, we would not be surprised to see CBOE become an acquisition target if the valuation continues to languish.



Modeling Assumptions

	2020	2021	2022	2023
Operating Revenue	3,224	2,941	3,146	3,334
Net Revenue	1,239	1,226	1,321	1,400
OpEx	541	554	559	565
EBIT	698	672	763	836
Net Income	488	473	539	592
Shares	110	106	103	99
Adj. EPS*	5.25	5.04	5.85	6.60
Tgt Multiple (Trailing PE)	22x	22x	22x	22x
Fair Value Est.	115	111	129	145
% Up/(Down)	29%	25%	45%	63%

*Excludes amortization of intangible assets

Risks

- Lower trading volume in exclusive and/or proprietary products, including VIX-related ETP products
- Increased competitive intensity, particularly in cash equities and multiply-listed options
- Regulation (e.g. data pricing, transaction taxes)

I do not hold a position with the issuer such as employment, directorship, or consultancy.

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Catalyst

- More robust trading activity
- M&A (acquirer / target)
- Successful new product launches
- Growing contribution from non-transaction revenue

Messages

Subject	Cash equities earnings
Entry	10/07/2020 12:53 PM
Member	agape1095
What's your expectation for cash equities earnings going forward and the impact of MEMX?	
Subject	Re: Cash equities earnings
Entry	10/07/2020 01:51 PM
Member	GLSV
TBD since it just launched. We should have a better idea whether they can get traction in the next few months. Clearly has the potential to be negative for pricing; however, we think of cash equities as a normalized low double digit contributor to profit (more this year obviously) and which should decline in importance over time.	
Subject	Author Exit Recommendation
Entry	06/28/2021 10:02 AM
Member	GLSV
The author has recommended exiting the position	
Subject	Re: Author Exit Recommendation
Entry	06/28/2021 08:58 PM
Member	Alpinist
GLSV, any color on the reason(s) for exiting the position? Thank in advance.	
Subject	Re: Re: Author Exit Recommendation
Entry	08/18/2021 01:35 PM
Member	GLSV
Nothing in particular. Was just thinking the risk/reward was more balanced absent a takeout (which was part of the original thesis)....and which just arrived today. Palm to forehead!	
Subject	Re: Re: Re: Author Exit Recommendation
Entry	08/19/2021 11:13 AM
Member	GLSV
Even though CME denying, FT is usually pretty good. Wouldn't be surprised if it was Sprecher @ ICE (aggressive), and there are other potential buyers.	